Trigonometry & Additional Topics

ANSWER KEY

Section 1: Trigonometric Ratios & Cofunctions

1 If $\sin(x^\circ) = \cos(42^\circ)$, where $0 < x < 90$, what is the value of x?

a. 42

b. 48 ✓

c. 52

d. 138

By the cofunction identity, $\sin(x^\circ) = \cos(90^\circ - x^\circ)$. Thus $90 - x = 42 \Rightarrow x = 48$.

2 In right triangle ABC with right angle at C, if $\cos A = \frac{5}{13}$, what is the value of $\tan A$?

a. $\frac{5}{12}$

b. $\frac{12}{5}$ ✓

c. $\frac{12}{13}$

d. $\frac{13}{12}$

Adjacent $= 5$, hypotenuse $= 13$, opposite leg $= \sqrt{13^2 - 5^2} = 12$. Therefore, $\tan A = \frac{12}{5}$.

3 Which trigonometric ratio is equal to $\frac{1}{\cos\theta}$?

a. $\sin\theta$

b. $\cot\theta$

c. $\sec\theta$ ✓

d. $\csc\theta$

By reciprocal definition: $\sec\theta = \frac{1}{\cos\theta}$.

Section 2: Unit Circle & Radians

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 Convert 210° to radians.

a. $\frac{5\pi}{6}$

b. $\frac{7\pi}{6}$ ✓

c. $\frac{4\pi}{3}$

d. $\frac{11\pi}{6}$

Multiply by $\frac{\pi}{180^\circ}$: $210 \times \frac{\pi}{180} = \frac{7\pi}{6}$.

5 What is the exact value of $\sin(\frac{4\pi}{3})$?

a. $-\frac{\sqrt{3}}{2}$ ✓

b. $\frac{\sqrt{3}}{2}$

c. $-\frac{1}{2}$

d. $\frac{1}{2}$

$\frac{4\pi}{3}$ is in Quadrant III where sine is negative. Reference angle is $\frac{\pi}{3}$. Thus $\sin(\frac{4\pi}{3}) = -\sin(\frac{\pi}{3}) = -\frac{\sqrt{3}}{2}$.

6 If θ is in Quadrant IV and $\cos\theta = \frac{3}{5}$, what is $\sin\theta$?

a. $-\frac{4}{5}$ ✓

b. $\frac{4}{5}$

c. $-\frac{3}{4}$

d. $\frac{3}{4}$

In Quadrant IV, sine is negative: $\sin\theta = -\sqrt{1-\cos^2\theta} = -\sqrt{1-\frac{9}{25}} = -\frac{4}{5}$.

Section 3: Complex Numbers & Arithmetic

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 What is the value of $(3+4i)(2-i)$ where $i^2=-1$?

a. $10+5i$ ✓

b. $2+5i$

c. $6-4i$

d. $10-5i$

Expand: $3(2)-3i+8i-4i^2=6+5i-4(-1)=6+5i+4=10+5i$.

8 What is the simplified form of $\frac{10}{3-i}$?

a. $3+i$ ✓

b. $3-i$

c. $\frac{3+i}{2}$

d. $3+2i$

Multiply by conjugate $(3+i)$: $\frac{10(3+i)}{(3-i)(3+i)}=\frac{10(3+i)}{9-i^2}=\frac{10(3+i)}{10}=3+i$.

9 What is the value of i^{102}?

a. 1

b. -1 ✓

c. i

d. $-i$

Divide power by 4: $102=4(25)+2$. Thus $i^{102}=(i^4)^{25}\cdot i^2=(1)^{25}(-1)=-1$.

Section 4: Coordinate Geometry of Circles

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 What is the radius of the circle defined by $(x+4)^2+(y-7)^2=81$?

a. 7

b. 9 ✓

c. 18

d. 81

In $(x-h)^2+(y-k)^2=r^2$, $r=\sqrt{81}=9$.

11 What is the center of the circle $x^2+y^2-10x+6y+9=0$?

a. $(5,\ -3)$ ✓

b. $(-5, 3)$

c. $(10,\ -6)$

d. $(-10, 6)$

Complete the square: $(x^2-10x+25)+(y^2+6y+9)=-9+25+9 \Rightarrow (x-5)^2+(y+3)^2=25$. Center is $(5,\ -3)$.

12 Which of the following points lies on the circle $(x-1)^2+(y-2)^2=25$?

a. $(4, 6)$ ✓

b. $(5, 5)$

c. $(1, 7)$

d. $(6, 2)$

Test $(4, 6)$: $(4-1)^2+(6-2)^2=3^2+4^2=9+16=25$ (true).

© ClipSAT: By Mr. Mohamed Abdallah +966597688647